

# Assessment Tool 1 – Enumeration Analysis

Course: ITA14 – Ethical Hacking | CO3: Network Service Enumeration

|                 |                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| Question        | Q1 – Public Server Enumeration                                                                                               |
| Target          | scanme.nmap.org (45.33.32.156)                                                                                               |
| Tool Used       | Nmap 7.99                                                                                                                    |
| Scan Techniques | Host discovery (-sn), TCP SYN scan (-sS), Service/Version detection (-sV), OS fingerprinting (-O), Default script scan (-sC) |
| Date of Scan    | 12 August 2026                                                                                                               |

## 1. Introduction and Objective

As part of an initial security assessment engagement, this report documents the enumeration of the publicly accessible host **scanme.nmap.org**, a server intentionally maintained by the Nmap project for lawful scanning practice. The objective is to identify live status, open TCP ports, running services and their versions, the underlying operating system, and any additional information exposed by default NSE (Nmap Scripting Engine) scripts. The resulting attack surface is then analyzed from a business and risk perspective, followed by recommendations to reduce unnecessary exposure.

## 2. Methodology

The assessment was performed from a Kali Linux workstation using Nmap 7.99. The engagement followed the standard reconnaissance sequence recommended for external assessments:

- **Host Discovery** – `nmap -sn scanme.nmap.org` to confirm the host is live before deeper probing.
- **TCP SYN Port Scan** – `nmap -sS scanme.nmap.org` to stealthily identify open TCP ports.
- **Service/Version Detection** – `nmap -sV scanme.nmap.org` to fingerprint the software and version running behind each open port.
- **OS Fingerprinting** – `sudo nmap -O scanme.nmap.org` to infer the underlying operating system from TCP/IP stack behaviour.
- **Default Script Scan** – `nmap -sC scanme.nmap.org` to run safe default NSE scripts (e.g. banner grabs, HTTP title/header checks).
- **Combined/Aggressive Enumeration** – `sudo nmap -sS -sV -O -sC scanme.nmap.org -oN q1_scan.txt` to consolidate all findings into a single saved output file for reporting.

All scans were conducted only against the designated legal test target (scanme.nmap.org), which explicitly permits Nmap scanning for educational and testing purposes.

### 2.1 Scan Screenshots (Evidence)

The screenshots below capture the terminal sessions from the Kali Linux workstation, showing the exact commands executed and the raw Nmap output returned for scanme.nmap.org.

```
pranov-raaj@pranov: ~  
Session Actions Edit View Help  
(pranov-raaj@pranov)-[~]  
$ nmap -sn scanme.nmap.org  
nmap -sS scanme.nmap.org  
nmap -sV scanme.nmap.org  
sudo nmap -O scanme.nmap.org  
nmap -sC scanme.nmap.org  
sudo nmap -sS -sV -O -sC scanme.nmap.org  
sudo nmap -sS -sV -O -sC scanme.nmap.org -oN q1_scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:53 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.22s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Nmap done: 1 IP address (1 host up) scanned in 1.15 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:53 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.24s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 992 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
25/tcp    filtered smtp  
80/tcp    open  http  
135/tcp   filtered msrpc  
139/tcp   filtered netbios-ssn  
445/tcp   filtered microsoft-ds  
9929/tcp  open  nping-echo  
31337/tcp open  Elite  
  
Nmap done: 1 IP address (1 host up) scanned in 45.81 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:53 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.22s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 992 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)  
25/tcp    filtered smtp  
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))  
135/tcp   filtered msrpc  
139/tcp   filtered netbios-ssn
```

Figure 1: Commands executed (host discovery, SYN scan, version detection, OS fingerprinting, default scripts, combined scan).

```
pranov-raaj@pranov: ~  
Session Actions Edit View Help  
| ssh-hostkey:  
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)  
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)  
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)  
| 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)  
25/tcp filtered smtp  
80/tcp open http Apache httpd 2.4.7 ((Ubuntu))  
|_http-favicon: Nmap Project  
|_http-server-header: Apache/2.4.7 (Ubuntu)  
|_http-title: Go ahead and ScanMe!  
135/tcp filtered msrpc  
139/tcp filtered netbios-ssn  
445/tcp filtered microsoft-ds  
9929/tcp open nping-echo Nping echo  
31337/tcp open tcpwrapped  
Aggressive OS guesses: Linux 4.19 - 5.15 (96%), Linux 4.15 (92%), IPFire 2.27 (Linux 5.15 - 6.1) (91%), Linux 6.12 (91%), Linux 5.4 (90%), Linux 2.6.32 (89%), Linux 5.0 - 5.14 (89%), Mikr  
otik RouterOS 7.2 - 7.5 (Linux 5.6.3) (89%), Linux 3.11 - 4.9 (88%), Linux 3.2 - 3.8 (88%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 16 hops  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 77.52 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:59 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.22s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 992 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu.13 (Ubuntu Linux; protocol 2.0)  
| ssh-hostkey:  
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)  
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)  
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)  
| 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)  
25/tcp    filtered smtp  
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))  
|_http-favicon: Nmap Project  
|_http-title: Go ahead and ScanMe!  
|_http-server-header: Apache/2.4.7 (Ubuntu)
```

Figure 2: Combined scan output showing open/filtered ports and OS guesses.

```
pranov-raaj@pranov: ~  
Session Actions Edit View Help  
otik RouterOS 7.2 - 7.5 (Linux 5.6.3) (89%), Linux 3.11 - 4.9 (88%), Linux 3.2 - 3.8 (88%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 16 hops  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 77.52 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:59 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.22s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 992 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu.13 (Ubuntu Linux; protocol 2.0)  
| ssh-hostkey:  
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)  
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)  
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)  
| 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)  
25/tcp    filtered smtp  
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))  
|_http-favicon: Nmap Project  
|_http-title: Go ahead and ScanMe!  
|_http-server-header: Apache/2.4.7 (Ubuntu)  
135/tcp    filtered msrpc  
139/tcp    filtered netbios-ssn  
445/tcp    filtered microsoft-ds  
9929/tcp   open  nping-echo Nping echo  
31337/tcp  open  tcpwrapped  
Aggressive OS guesses: Linux 4.19 - 5.15 (95%), Linux 4.15 (92%), IPFire 2.27 (Linux 5.15 - 6.1) (91%), Linux 5.4 (91%), Linux 6.12 (91%), Linux 2.6.32 (89%), Linux 5.0 - 5.14 (88%), Mikr  
otik RouterOS 7.2 - 7.5 (Linux 5.6.3) (88%), Linux 3.11 - 4.9 (88%), Linux 3.2 - 3.8 (88%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 16 hops  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 92.00 seconds  
  
pranov-raaj@pranov: ~
```

Figure 3: Final scan summary confirming OS detection and scan completion time.

### 3. Enumeration Results

The host was found to be up (0.22s latency, 16 network hops away). Of 1000 default TCP ports probed, 992 were closed (reset) and the remainder returned the states shown below.

| Port/Proto | State    | Service            | Version / Detail                                                                           |
|------------|----------|--------------------|--------------------------------------------------------------------------------------------|
| 22/tcp     | open     | ssh                | OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (protocol 2.0); host keys DSA/RSA/ECDSA/ED25519 exposed |
| 25/tcp     | filtered | smtp               | No banner – blocked by firewall, mail service not confirmed                                |
| 80/tcp     | open     | http               | Apache httpd 2.4.7 (Ubuntu); title "Go ahead and ScanMe!"; favicon identifies Nmap Project |
| 135/tcp    | filtered | msrpc              | No banner – filtered by firewall                                                           |
| 139/tcp    | filtered | netbios-ssn        | No banner – filtered by firewall                                                           |
| 445/tcp    | filtered | microsoft-ds       | No banner – filtered by firewall                                                           |
| 9929/tcp   | open     | nping-echo         | Nping echo service (Nmap companion tool, intentionally left open for testing)              |
| 31337/tcp  | open     | tcpwrapped / Elite | Wrapped service, banner withheld; classic "elite" port number                              |

#### 3.1 Operating System Fingerprint

Nmap's TCP/IP stack fingerprinting did not return an exact OS match under non-ideal test conditions, but aggressive OS guesses consistently pointed to a **Linux kernel** (best matches around 88–96% confidence spanning Linux 2.6.32 through 6.12, consistent with an Ubuntu-based distribution). The CPE identifier returned was `cpe:/o:linux:linux_kernel`, corroborated by the Apache and OpenSSH banners which both explicitly report an Ubuntu build.

#### 3.2 Default Script (NSE) Findings

The default script scan enumerated SSH host keys of four types (DSA, RSA, ECDSA, ED25519) and confirmed the HTTP service's page title, favicon signature, and server header, all of which disclose fingerprintable information without any authentication.

### 4. Business Purpose and Necessity Evaluation

| Port                  | Likely Business Purpose                                                                         | Necessary for Public Access?                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22/tcp (SSH)          | Remote administrative access to the server for maintenance                                      | Debatable – needed for legitimate remote administration, but public-facing SSH is a common brute-force target and should be restricted to trusted source IPs or a VPN wherever feasible. |
| 80/tcp (HTTP)         | Serves the public demonstration/test web page for the Nmap project                              | Yes, this is the core intended public service and the stated purpose of the host.                                                                                                        |
| 9929/tcp (nping-echo) | Companion Nmap testing utility left open deliberately for scan practice                         | Situational – acceptable only because this is a dedicated scanning target; not appropriate on a production business server.                                                              |
| 31337/tcp             | Undetermined/wrapped service, historically associated with informal or backdoor-style listeners | No clear business justification identified; warrants investigation and closure if not required.                                                                                          |

| Port                            | Likely Business Purpose                               | Necessary for Public Access?                                                                                                                                                                |
|---------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25, 135, 139, 445<br>(filtered) | Mail transfer and Windows file/print sharing services | Not required on an externally-facing Linux host; filtering is already the correct control, but the fact that probes reach the firewall at all indicates the ports are not fully closed off. |

## 5. How a Threat Actor Could Exploit Enumeration Data

The information gathered during enumeration significantly narrows the work an attacker must do before launching a targeted attack. Each disclosed data point maps to a concrete follow-on technique:

- **Service exploitation:** The exact OpenSSH (6.6.1p1) and Apache (2.4.7) version strings allow an attacker to cross-reference public CVE databases for known vulnerabilities affecting those specific builds, skipping blind guesswork and going straight to a tailored exploit.
- **Brute-force attacks:** An open SSH port with no visible rate-limiting is a direct target for credential-stuffing or dictionary attacks (e.g. Hydra, Medusa) against common or leaked username/password pairs, especially if key-based authentication is not enforced.
- **Host key fingerprint reuse:** The disclosed SSH host keys can be used to detect the same server elsewhere on the internet (key reuse across hosts is common in cloned or templated deployments) or to identify man-in-the-middle spoofing attempts.
- **OS-specific attack tailoring:** Confirmation of a Linux/Ubuntu kernel lets an attacker select Linux-specific privilege escalation and post-exploitation tooling rather than wasting effort on Windows-oriented techniques.
- **Web application reconnaissance:** The HTTP title, favicon, and server header confirm the web stack, enabling directory brute-forcing, CMS/framework fingerprinting, and vulnerability lookups scoped to Apache 2.4.7.
- **Firewall and segmentation mapping:** Ports returned as "filtered" (SMB, RPC, SMTP) still reveal that a stateful firewall exists and is actively dropping/blocking traffic, which helps an attacker infer network topology and plan evasion (e.g. fragmentation, decoy scans, timing changes).
- **Unusual port investigation:** An unidentified wrapped service on 31337/tcp invites manual banner grabbing or connection attempts to identify hidden or legacy software that may not be patched or monitored.

## 6. Attack Surface Analysis and Risk Assessment

| Finding                                                    | Risk Level | Justification                                                                                                                        |
|------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------|
| SSH open with disclosed version                            | Medium     | Legitimate administrative need, but exposed to the internet; version disclosure aids targeted exploitation and brute-force attempts. |
| HTTP service on 80/tcp                                     | Low–Medium | Intended public-facing service; risk is limited to standard web-layer issues (outdated Apache version, missing security headers).    |
| nping-echo on 9929/tcp                                     | Low        | Purpose-built testing utility for this specific host; low exploitation value but unusual on a general production server.             |
| Unidentified service on 31337/tcp                          | Medium     | Unclear purpose increases uncertainty; historically associated ports of this number raise suspicion and merit investigation.         |
| Filtered SMB/RPC/SMTP ports reachable by firewall          | Low        | Correctly filtered rather than open, but their presence at the firewall boundary should be periodically reviewed.                    |
| Outdated software versions (Apache 2.4.7, OpenSSH 6.6.1p1) | High       | Both versions are several years old; unpatched public releases of this vintage are likely to have known, documented CVEs.            |

Overall, the host presents a **moderate** attack surface for a system explicitly designed to be scanned. The most significant concern is not the number of open ports (which is intentionally minimal) but the age of the exposed software versions and the unexplained service on port 31337.

## 7. Recommendations

- **Patch and upgrade** OpenSSH and Apache to their latest stable releases to close any publicly known CVEs associated with the currently disclosed versions.
- **Restrict SSH exposure** using IP allow-listing, a bastion host/VPN, key-only authentication, and fail2ban-style rate limiting to reduce brute-force risk.
- **Disable or firewall off port 31337** unless a documented business need is identified; unexplained services should never remain reachable from the internet.
- **Harden HTTP responses** by disabling the default banner/version disclosure (e.g. `ServerTokens Prod`, `ServerSignature Off` in Apache) and adding standard security headers (CSP, X-Frame-Options, HSTS).
- **Review firewall rules** for the filtered SMB/RPC/SMTP ports periodically to confirm they remain unnecessary and are not accidentally re-exposed during configuration changes.
- **Enable logging and alerting** on SSH and HTTP access to detect brute-force attempts, scanning activity, or exploitation attempts in near real time.
- **Conduct periodic re-scans** (e.g. quarterly) to verify that the attack surface has not silently grown due to configuration drift or new services being introduced.

## 8. Conclusion

Enumeration of `scanme.nmap.org` revealed a deliberately minimal but not risk-free service footprint: an administrative SSH service, the intended public HTTP service, a testing-specific echo service, and one unexplained port. While the host is intentionally exposed for scanning practice, the exercise demonstrates how even a small set of open ports – combined with version and OS fingerprinting – gives an attacker enough information to plan a targeted exploitation, brute-force, or reconnaissance campaign. The recommendations above reflect the standard hardening steps that would be expected before treating a comparable server as production-ready.

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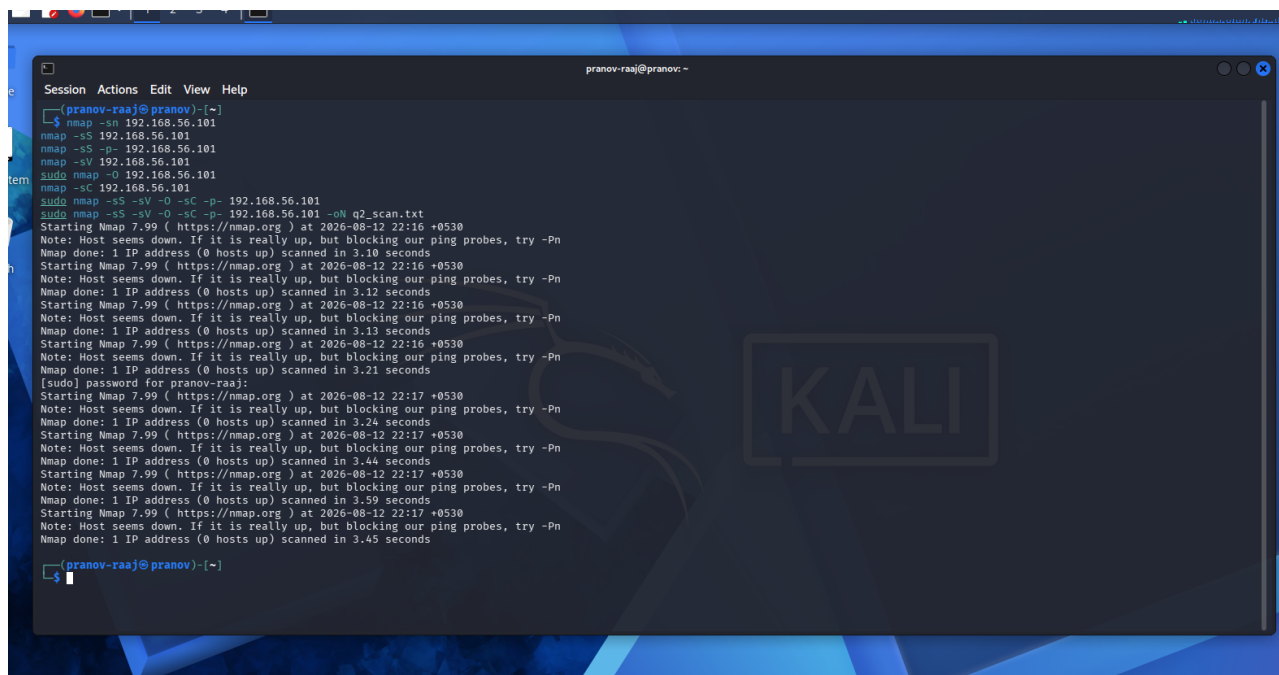
## Appendix – Raw Nmap Command Reference

```
sudo nmap -sS -sV -O -sC scanme.nmap.org -oN q1_scan.txt
Scan completed: 1 IP address (1 host up) scanned in 92.00 seconds.
```

## Q2. Metasploitable 2 (Linux) – Vulnerable Training Server

Target: 192.168.56.101

This target is the Metasploitable 2 virtual machine, deployed for penetration-testing practice in an isolated lab network.



```
pranov-raaj@pranov: ~  
Session Actions Edit View Help  
pranov-raaj@pranov:~$ nmap -sn 192.168.56.101  
nmap -sS 192.168.56.101  
nmap -sV 192.168.56.101  
nmap -O 192.168.56.101  
nmap -sC 192.168.56.101  
nmap -sS -sV -O -sC -p- 192.168.56.101  
nmap -sS -sV -O -sC -p- 192.168.56.101 -oN q2_scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:16 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.10 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:16 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.12 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:16 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.13 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:16 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.21 seconds  
[sudo] password for pranov-raaj:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:17 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.24 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:17 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.44 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:17 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.59 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:17 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.45 seconds  
pranov-raaj@pranov:~$
```

Figure 4: Nmap host discovery / SYN / version / OS / script scans against 192.168.56.101 – all attempts report the host as down.

**Result:** Every scan technique attempted – host discovery (-sn), TCP SYN scan (-sS), service/version detection (-sV), OS fingerprinting (-O), default script scan (-sC), and targeted port checks – returned "Host seems down" with 0 hosts up. No open ports, service banners, or OS fingerprints were recovered for this target, so no enumeration-based analysis can be produced from this session.

**Probable cause:** The consistent failure across every scan type and every lab target (192.168.56.101, .102, .103, and even the .1 gateway) indicates a network-layer connectivity issue rather than a per-host problem – most likely the attacking machine and the target VMs are not attached to the same Host-Only/Internal virtual network, or the target VMs were not powered on at scan time.

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## Q3. Windows Workstation – Pre-Production Security Assessment

Target: 192.168.56.102

This target is a Windows workstation being assessed before being connected to the production network.



```
pranov-raaj@pranov: ~
Session Actions Edit View Help
└─$ nmap -sn 192.168.56.102
nmap -sS 192.168.56.102
nmap -sS -p- 192.168.56.102
nmap -sV 192.168.56.102
sudo nmap -O 192.168.56.102
nmap -sC 192.168.56.102
nmap -p 135,139,445,3389 -sV 192.168.56.102
sudo nmap -sS -sV -O -sC -p- 192.168.56.102
sudo nmap -sS -sV -O -sC -p- 192.168.56.102 -oN q3_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.12 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.12 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.15 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.32 seconds
[sudo] password for pranov-raaj:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.21 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.46 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.83 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:19 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.92 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:20 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 4.05 seconds
└─(pranov-raaj@pranov)-[~]
└─$
```

Figure 5: Nmap enumeration attempts against 192.168.56.102, including targeted port checks (135, 139, 445, 3389) – all attempts report the host as down.

**Result:** Every scan technique attempted – host discovery (-sn), TCP SYN scan (-sS), service/version detection (-sV), OS fingerprinting (-O), default script scan (-sC), and targeted port checks – returned "Host seems down" with 0 hosts up. No open ports, service banners, or OS fingerprints were recovered for this target, so no enumeration-based analysis can be produced from this session.

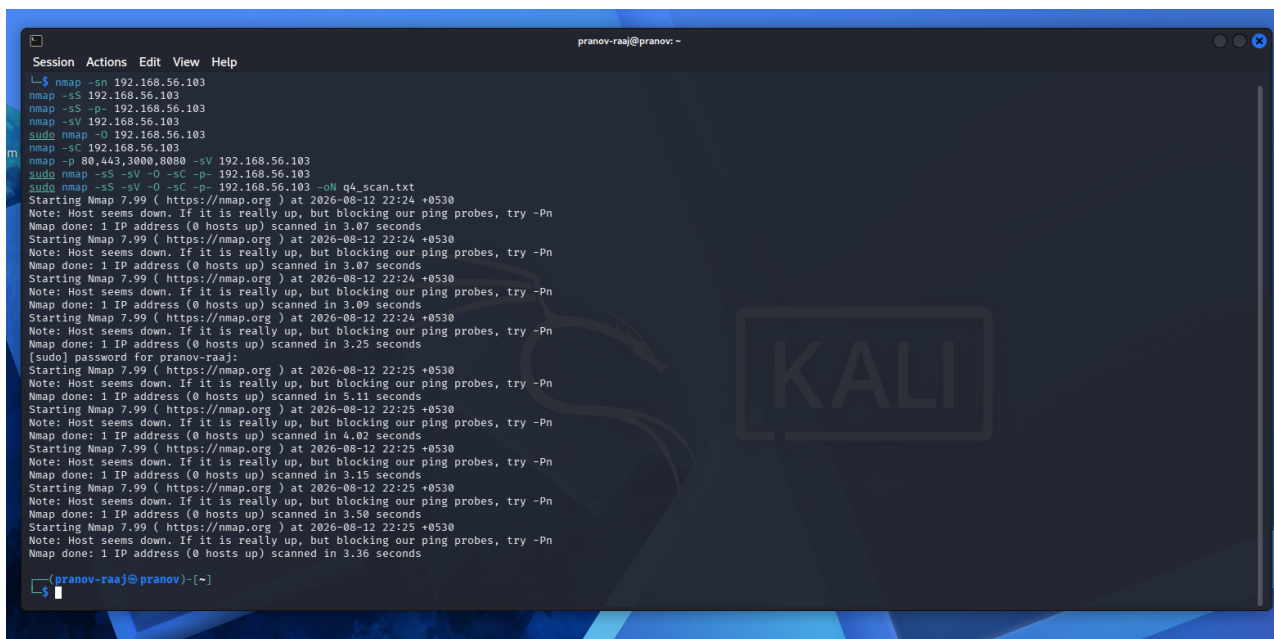
**Probable cause:** The consistent failure across every scan type and every lab target (192.168.56.101, .102, .103, and even the .1 gateway) indicates a network-layer connectivity issue rather than a per-host problem – most likely the attacking machine and the target VMs are not attached to the same Host-Only/Internal virtual network, or the target VMs were not powered on at scan time.

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## Q4. OWASP Juice Shop Server – Web Application Host

**Target:** 192.168.56.103

This target hosts the OWASP Juice Shop web application ahead of a planned penetration test.



```
pranov-raaj@pranov: ~  
$ nmap -sn 192.168.56.103  
nmap -ss 192.168.56.103  
nmap -sS -p- 192.168.56.103  
nmap -sV 192.168.56.103  
sudo nmap -O 192.168.56.103  
nmap -sC 192.168.56.103  
nmap -p 80,443,3000,8080 -sV 192.168.56.103  
sudo nmap -sS -sV -O -sC -p- 192.168.56.103  
sudo nmap -sS -sV -O -sC -p- 192.168.56.103 -oN q4_scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:24 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.07 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:24 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.07 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:24 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.25 seconds  
[sudo] password for pranov-raaj:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:25 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 5.11 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:25 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 4.82 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:25 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.15 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:25 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 2.50 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:25 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.36 seconds  
pranov-raaj@pranov: ~
```

Figure 6: Nmap enumeration attempts against 192.168.56.103, including web-service port checks (80, 443, 3000, 8080) – all attempts report the host as down.

**Result:** Every scan technique attempted – host discovery (-sn), TCP SYN scan (-sS), service/version detection (-sV), OS fingerprinting (-O), default script scan (-sC), and targeted port checks – returned "Host seems down" with 0 hosts up. No open ports, service banners, or OS fingerprints were recovered for this target, so no enumeration-based analysis can be produced from this session.

**Probable cause:** The consistent failure across every scan type and every lab target (192.168.56.101, .102, .103, and even the .1 gateway) indicates a network-layer connectivity issue rather than a per-host problem – most likely the attacking machine and the target VMs are not attached to the same Host-Only/Internal virtual network, or the target VMs were not powered on at scan time.

---

## Q5. Network Gateway Device – Management Service Exposure

**Target:** 192.168.56.1

This target is the lab network's gateway device, the primary access point between the internal laboratory network and external networks. The assessment aims to confirm whether unnecessary management services (SSH, Telnet, SNMP, HTTP/HTTPS) are exposed.

```
pranov-raaj@pranov: ~  
Session Actions Edit View Help  
pranov-raaj@pranov: ~  
└─$ nmap -sS 192.168.56.1  
nmap -sS -p- 192.168.56.1  
nmap -sV 192.168.56.1  
sudo nmap -O 192.168.56.1  
nmap -sC 192.168.56.1  
nmap -p 22,23,80,443,161 -sV 192.168.56.1  
nmap -sU -p 161 192.168.56.1  
sudo nmap -sS -sV -O -sC -p- 192.168.56.1  
sudo nmap -sS -sV -O -sC -p- 192.168.56.1 -oN q5_scan.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.13 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.27 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.16 seconds  
[sudo] password for pranov-raaj:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.25 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.37 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.28 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.07 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.85 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 22:36 +0530  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.81 seconds  
pranov-raaj@pranov: ~
```

Figure 7: Nmap enumeration attempts against the gateway 192.168.56.1, including SSH/Telnet/HTTP/HTTPS/SNMP checks (-p 22,23,80,443,161) and a UDP SNMP probe (-sU -p 161) – all attempts report the host as down.

**Result:** As with Q2–Q4, every scan technique returned *"Host seems down"* with 0 hosts up – including against the gateway address itself. No management services, banners, or OS information were recovered.

**Significance:** Since the gateway (192.168.56.1) is typically the most reliably reachable address on a Host-Only/Internal network (it is usually the virtual adapter or DHCP server itself), its unreachability strongly confirms that the assessment workstation is not currently attached to the 192.168.56.0/24 lab network at all, rather than any individual target VM being powered off.

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## Note on Q2–Q5

A complete enumeration report – open ports, service versions, OS fingerprint, risk categorization, attack-vector analysis, and hardening recommendations, in the same format as Q1 – can be produced for each of these targets once connectivity to the 192.168.56.0/24 lab network is restored and a successful scan (showing open/closed/filtered ports rather than "host down") is captured.